Edutechlife



MODULE 3 · GEMINI

Gemini in Practice

Google's AI ecosystem, Deep Research and workflows for investigating with verifiable sources

Module 3
IALab · Edutechlife

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Teaching guide

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Gemini is Google's family of models for conversation, analysis and research. Its biggest advantage is that it **lives where you already work**: Gmail, Docs, Drive and Search. Here you will learn to use it for research without letting it invent data.

5
products in the ecosystem

4
steps in Deep Research

3
sources needed before you conclude

01 What Gemini is today

Gemini is developed by Google DeepMind and is available in the Gemini app, in Google Search and inside Google Workspace. Google updates its models often, so the version you use depends on your account.

CHECK YOUR VERSION

Open the model selector in your account and note which version you have. If a tutorial mentions a different one, it has probably changed.

02 The ecosystem you already use

Product	What it adds	Example of use
Gmail	Summaries and drafts of long threads	Summarize 30 emails from a supplier and list the open commitments.
Docs	Drafting, rewriting and tone review	Turn your class notes into a text with an introduction and a conclusion.
Drive	Searching and summarizing your own files	Find where you mentioned a budget across your documents from this year.
Sheets	Analyzing tables and suggesting formulas	Calculate the average per course and flag unusual values.
Search	Answers with links to sources	Compare two rules or prices and review the sites it cites.

03 Deep Research: multi-step investigation

Deep Research explores dozens of sources, synthesizes them and delivers a report. It is powerful, but **you define the question and verify the result**.



Deep Research explores sources; you review the plan and the report.

1. **Define the question:** topic, audience, time frame and what to exclude.
2. **Review the plan:** the researcher proposes its search lines; adjust what is missing or unnecessary.
3. **Read the report with its sources:** open at least three sources that support the key conclusions.
4. **Ask for the limitations:** what could it not verify? Do not use that part without further review.

Worked example: a market report

Example: investigate demand for AI courses for adults in Colombia.

PROMPT

```
Investigate demand for AI courses for adults in Colombia over the last two
years.
Audience: the director of a continuing-education institution.
Deliver: 1) a five-line executive summary, 2) five findings with source and
date, 3) figures that conflict between sources, 4) what you could not verify.
```

BEFORE YOU USE THE REPORT

Open three sources that support its main conclusions. If a figure does not appear in the source it cites, discard it.

04 Investigating without invented facts

A good research workflow separates three things: **facts with a source**, **interpretations** and **assumptions**. Ask for that separation explicitly.

WARNING SIGNS

- A figure with no source, or a source that does not contain it.
- A name or quote you cannot find when you search for it.
- An answer that sounds very certain about a recent topic.

05 Multimodal: text, images, audio and video

Gemini can analyze documents, images, charts, audio and video. Use it to extract information from a chart, transcribe a recording or compare two documents. Check the size and format limits of your plan.



Multimodal models combine text, images and audio in a single workflow.

06 A research prompt

PROMPT

```
Investigate [topic] for [audience].
Look for sources from the last [years] years from official bodies, universities
or recognized media.
```

```
Deliver: 1) a five-line summary, 2) findings with their source, 3)
contradictions between sources.
Cite every claim with its source and separate facts from interpretations.
State what you could not verify.
```

07 Privacy and permissions

Before you connect Gmail, Drive or Docs, check what permissions it asks for and grant only those you need. Do not upload personal data about others, medical records, minors' data or financial information without authorization. In Colombia, **Law 1581 of 2012** applies.

08 Common mistakes

Mistake	What happens	How to avoid it
Accepting the first result	Unverified data gets used	Open and check the key sources
Questions that are too broad	Generic answers	Narrow the topic, audience and period
Trusting the summary, not the original	Nuance gets lost	Read the part of the original that supports each conclusion
Uploading sensitive data	Privacy risk	Anonymize before uploading

09 Worked example: grading a report

Use this rubric to decide whether a report is ready for a decision. Score each criterion as strong, partial or insufficient.

Criterion	Strong	Partial	Insufficient
Sources	Every finding has a verifiable source and date	Some sources without a date	Claims with no source
Accuracy	Figures match the original source	Rounded figures without explanation	Figures that are not in the source
Nuance	Points out contradictions and conditions	Presents only one version	Presents the conclusion as final

Criterion	Strong	Partial	Insufficient
Limitations	States what it could not verify	Mentions limits in general terms	States no limits

10 Field exercises

1. **Local map:** research three competitors in your city using public sources. Deliver a table with price, offer and source.
2. **Verification:** take a news item that reached you on WhatsApp and check it with Search. Note which sources confirm or contradict it.
3. **Report with Deep Research:** research a topic from your field, verify three sources and rewrite the report in your own words.

TO WRAP UP

- [ ] I define the question, audience and scope before I investigate.
- [ ] I verify at least three sources before I use a conclusion.
- [ ] I separate facts, interpretations and assumptions.
- [ ] I check permissions and do not share sensitive data about others.